

Scylla's Data Centers; the Incredible Water Consumption of Data Centers in California, Why it Matters, and How to Manage It
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Abstract:

California's ability to secure its economic and sustainability interests in the expanding market of data centers depends on holistic legislation surrounding water usage by these corporations. California has a significant amount of leverage to create relationships with data centers on a local and state level that increase economic investment without losing out on sustainable water usage and public acceptance. Currently, data centers are a hot-button topic that face increased scrutiny from state residents due to a lack of transparency on water usage. A two-pronged approach focusing on mandating state-wide reporting while at the same time incentivizing data centers to co-invest in recycled water facilities as a water source will help build public confidence in data centers and ease economic growth. In the long-term, the co-investment from recycled water will allow California to stay ahead of worsening droughts that have been significantly impactful on the state in the last few decades. This report analyzes important data surrounding existing water-usage reporting and lays out how reporting consumption and investment in recycled water can be beneficial to public-well being and the environment.

Recycled Water Policy:

Data center water usage is the central point around which this report focuses on. Data centers use water mainly for cooling

servers. Data centers use a variety of cooling technologies that therefore use varying amounts of water. A data center may use anywhere between 18K gallons to 500K gallons per day based on size and efficiency. Annually, that could be anywhere between 6.6 million gallons to 182.5 million gallons. In terms of swimming pools, that's between 10 and 276 olympic pools. This is a huge range; it is for this reason that regulating data centers water usage is increasingly important for the protection of common resources as well as following state sustainability guidelines. Per the UC Berkeley Center for Law, Energy, and the Environment's report titled *Regulating Data Center Water Usage in California*, data centers generally consume water from four sources: commercial water services, recycled water, surface water, and ground water. Each type is expanded below.

Commercial water services are commonplace for any private activity that uses large amounts of water. Water services specifically for data centers are not regulated by the federal or state government. Commercial water services are instead regulated as a whole, with California specifically regulating both private and public water providers holistically. A relationship between a data center and a municipal water provider may be created when data center water needs exceed the capacity of private vendors. When these relationships occur, it is important that the local authorities properly manage cost

increases and resource usage for the locals in order to prevent disputes over water rights.

Recycled water is another option, but is currently a marginal source of water provisions for data centers. Some data centers, when built near existing recycled water facilities, use it for the aforementioned cooling services. Recycled water facilities are generally smaller and more rare nationally, but California is strongly invested in increasing the number available for commercial services. In 2021 there were 267 facilities capable of producing Title 22 recycled water, which can be used for commercial cooling. The 2022 California Water Supply Strategy aims for producing 800K acre feet of recycled water by 2030 and 1.4 million acre feet by 2040. Recycled water is a growing sector of water services.

Both ground and surface water are very minimally regulated by California in terms of water usage. Commercial users are not regularly required to report their withdrawal rates. Groundwater and surface water are also mainly regulated on a local level, making state-wide discussion difficult and ineffective. This suggests that the most optimal method to holistically regulate data center water usage is to increase investment in recycled water.

Another reason for using recycled water is that data centers are as a whole, highly consumptive of the water used. Google reports that 79% of the water used in their data centers worldwide is consumed. Equinix states that 76% of water used by their data centers is consumed. While not inherently a positive or negative, water consumption is a useful metric to analyze

water usage. If an activity consumes more water than is passed on, it is likely better to use quality recycled water to prevent over consumption. Regardless, water that is not consumed is still discharged into municipal waste-water treatment as it needs to be treated before it can be returned to commercial use or to ground-water.

Due to these high water usage rates in data centers, the most equitable and economic system to incentivize data centers to build in California while keeping a focus on sustainable use of common resources is to promote a relationship with recycled water centers. Data centers, when investing in smaller municipalities should be incentivized to support expansion or development of recycled water centers. Large municipalities like Santa Clara and San Jose that are investing in data centers are already focused on this idea. Per the UC Berkeley report, both Santa Clara and San Jose require recycled water to be used when available. Both also offer discounted rates for water from recycled water facilities. Specifics were not available in the report, but it is an idea with a lot of potential.

One objection to pushing economic incentives is that data centers will choose to build in the least expensive regions. This may be true, however with increased investment into data centers across the country, as well as a strong concentration of tech companies in the silicon valley, California is in a unique position to leverage these advantages in order to expand the state's economy while at the same time supporting existing sustainability goals. Municipalities that are being looked at by

data-center companies should use their leverage for the good of their people.

Incentives for co-investment in recycled water facilities can include but are not limited to: lower rates on water from recycled water facilities, tax breaks when data centers explicitly invest in expanding recycled water facilities, and long-term relationships hinging on data center reliance on recycled water instead of commercial or in-ground sources. WUE (Water Usage Efficiency) rebates would also be a feasible economic tool to drive down consumption and support recycled water facility developments. Another interesting idea would be a water credit system similar to carbon credits. Due to a lack of water usage reporting, many of these ideas are still in the early stages of research.

That being said, recycled water initiatives are only part of the solution. Developing a comprehensive system of reporting and review is just as necessary to better prepare water systems to handle the needs of data centers.

Reporting Policy:

Different parts of California carry varying levels of water sensitivity. Some regions, such as the Bay Area and part of LA, are better equipped to handle the water demands of data centers. On the other hand, construction in regions such as Imperial County would put massive strain on a water system already heavily reliant on imports from the Colorado River.

A report by Next 10, *The Intersection of Data Center Development, Water Availability, and Environmental Justice in California*, found that this pattern

repeats across the state. The most troubling takeaway was this: we can never fully know the true impact. Outdated legislation makes it easy for contractors to avoid proper assessment.

Under the California Environmental Quality Act (CEQA), data centers can be built without disclosing the impacts they have on the surrounding community via dated zoning laws. Through a process known as ministerial approval and tiering, data centers can circumvent environmental reviews and other important requirements to streamline construction. Assembly Bill 93, which would have required water use reporting, was vetoed by Governor Gavin Newsom in 2025. As it currently stands, there is no clear path forward for getting more information to the public.

A big point of contention standing in the way of mandated reporting is the burdens it will place on data center constructors. "Trade secrets" may be on the line, and in an ultra-competitive space, no one is willing to give ground. This argument fails to hold water. Other industries are required to report environmental and public externalities they incur, such as greenhouse gas emissions from power plants or chemical disclosures from manufacturers. Since data centers are also highly impactful on common resources, they should be required to as well. Moreover, fostering an environment of public distrust is politically unwise for any incumbent. The biggest critique of data centers stems from public distrust of their water usage. The reality is that these data centers can be managed responsibly and have great potential for economic benefits, but it is only through

reporting and public education that these benefits can be fully realized.

Currently, a revised bill, Assembly Bill 2619, represents the most promising path forward for water reporting. It offers a strong model, mandating detailed disclosures on water consumption and requiring state agencies to develop guidelines and best practices by 2029, including the use of closed-loop cooling systems and non-potable water. The benefits of this bill far outweigh any potential costs that data center operators would incur. Santa Clara County, the only county in California where data centers actually disclose environmental information, shows that transparency is not only feasible but leads to better outcomes for communities and developers alike.

With reliable reporting, water providers can plan ahead for distribution, regulators can assess which areas are best suited for data centers and their water needs, and auditors can evaluate existing centers and identify opportunities for improvement. Most importantly, the public will be able to properly assess the costs and benefits of projects being built in their community.

Community Impact:

Within our own community, the Utilitarian foundation team visited a data center campus in Santa Clara. The same company, name redacted for safety purposes, also operates data centers in San Jose as well as Milpitas. Their centers are unassuming - the buildings are so plain as to not warrant a second look, and sit in the corner of a corporate park. The average person would never even know what these

buildings were if they drove past them. In this campus there are 7 separate data centers, which combined, based on square footage and average water usage of data centers, would likely use up to 2 million gallons of water a day. In other words, our estimate suggests that this campus uses approximately 746 million gallons of water a year. These 7 are only a small part of the existing 83 data centers in Santa Clara and an estimated 270 data centers in California. With some estimates of 60 more data centers in construction across California, data centers are important to be managed. As reservoirs run lower and lower, heat rises, and other environmental conditions magnify the finite nature of water reserves, the state's leaders must be proactive in managing this growing sector.

These data points prove that it is important that California uses economic and social incentives to push data center construction to co-invest in recycled water. It is also incredibly clear that a lack of data on water consumption is a glaring hole in government management of this sector.

Conclusion:

California has been at the center of technological advancements for the past few decades; therefore, it is in the state's best interest to proactively manage data centers to stay ahead of the curve. As the continued construction of data centers across the United States shines a spotlight on the issue, it is critical that California's legislature responds to public concern and works to proactively manage the way data centers are approved and environmentally reviewed. According to a variety of sources, water

security is the foremost concern surrounding data centers, and both reporting requirements and water optimization efforts are lackluster relative to the scale of the problem—and insufficiently supported by state policy.

As our paper shows, water security is the biggest concern around data centers. Both reporting and optimization efforts are currently inadequate and not properly supported by the government. We strongly support the passage of Assembly Bill 2619 in order to close loopholes for reporting and strengthen requirements to support water recycling initiatives. It is only through these

methods that California will ensure every citizen can reap the benefits of data centers while minimizing the strain on our shared water resources. California has a choice: we can continue to let data centers operate in the shadows, building public distrust and straining already fragile water systems, or we can lead the nation in creating a regulatory framework that balances innovation with sustainability. The path forward is clear. It is time for the legislature to act.

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